

Product Pricing Algorithm

Business Science

3/9/2026

Problem Statement

Research and Development wants help to determine new product ideas and pricing using existing product line as a benchmark.

Solution Summary

TODO: INSERT SOLUTION SUMMARY

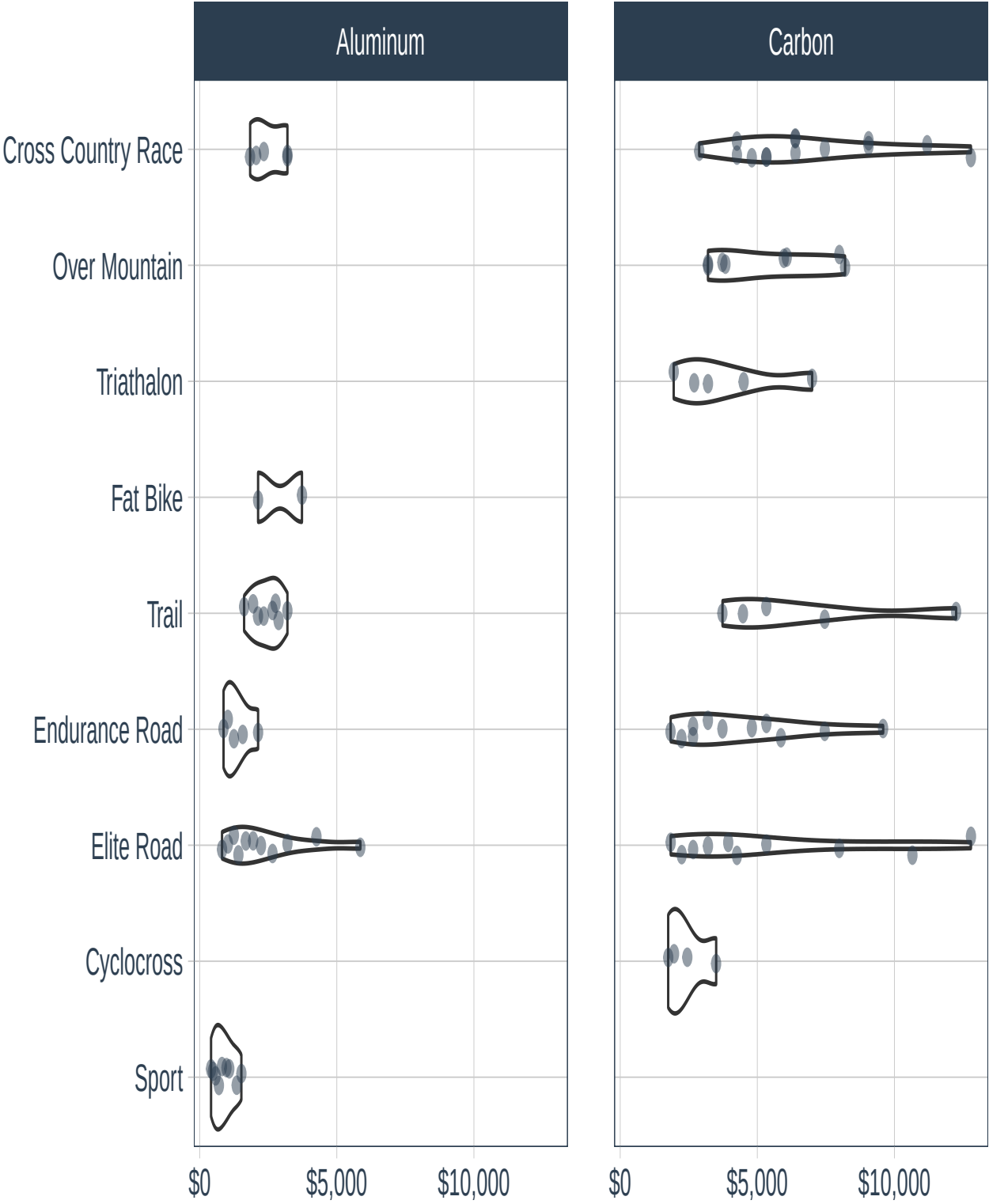
Gap Analysis

Bike List

TODO: INSERT BIKE LIST

```
## # A tibble: 97 x 15
##       id price model  category_1 category_2 frame_material base_model model_tier
##   <int> <dbl> <chr>  <chr>      <chr>      <chr>      <chr>      <chr>
## 1     1   6070 Jekyll Mountain Over Moun~ Carbon      Jekyll      Carbon 2
## 2     2   5970 Trigg~ Mountain Over Moun~ Carbon      Trigger     Carbon 2
## 3     3   2770 Beast~ Mountain Trail       Aluminum    Beast of ~ 1
## 4     4  10660 Super~ Road      Elite Road Carbon      Supersix ~ Hi-Mod Te~
## 5     5   3200 Jekyll~ Mountain Over Moun~ Carbon      Jekyll      Carbon 4
## 6     6  12790 Super~ Road      Elite Road Carbon      Supersix ~ Black Inc.
## 7     7   5330 Super~ Road      Elite Road Carbon      Supersix ~ Hi-Mod Du~
## 8     8   1570 Synap~ Road      Endurance~ Aluminum    Synapse     Disc 105
## 9     9   4800 Synap~ Road      Endurance~ Carbon      Synapse     Carbon Di~
## 10    10   480 Catal~ Mountain Sport       Aluminum    Catalyst     3
## # i 87 more rows
## # i 7 more variables: black <dbl>, hi_mod <dbl>, ultegra <dbl>, team <dbl>,
## #   red <dbl>, dura_ace <dbl>, disc <dbl>
```

Product Gap Analysis



Price Prediction

TODO: INSERT PRICE PREDICTION SUMMARY AND TABLE

New Model Attribute	Trigger Al 1	Slice Al 1
.pred	\$2,365	\$1,745
frame_material	Aluminum	Aluminum
category_2	Over Mountain	Triathalon
base_model	Trigger	Slice
model_tier	Aluminum 1	Ultegra
black	0	0
hi_mod	0	0
team	0	0
red	0	0
ultegra	0	0
dura_ace	0	0
disc	0	0